

SovereignGate

Enterprise Privacy Proxy & Multi-Agent Governance on Azure AI Foundry

- Built for regulated Contract Lifecycle Management (CLM) & SMEs
- Stops IP & PII leakage into untrusted model training loops
- Enforces regional data residency (UK DPA 2018 & EU GDPR) dynamically
- Seamless user experience: zero workflow friction, zero token leakage

THE PROBLEM

The Shadow AI Dilemma for SMEs

Balancing Frontier AI Productivity with Strict Regulatory Compliance

- **Uncontrolled Browser LLM Usage:** Employees paste sensitive client NDAs, master service agreements, and fee structures into free/public models.
- **Data Residency Violations:** Cross-border transmission breaches UK GDPR and Data Protection Act rules when routed through untrusted jurisdictions.
- **IP Poisoning:** Proprietary trade terms and party identities become permanent parts of public model re-training corpuses.
- **Legacy DLP Failure:** Traditional network firewalls simply block access, halting AI productivity and driving users toward riskier workarounds.

THE SOLUTION

The 3-Agent Sovereign Architecture

Decoupled Ingestion, Compliant Dispatch, and Ephemeral Reconstruction

- Agent 1 (Compliance Sentinel): Ingests raw contract text, redacts sensitive business PII/IP into deterministic tokens (e.g., {{ORG_1}}, {{MONEY_2}}), and classifies residency rules.
- Ephemeral Session Vault: Secures un-redacted values strictly in local volatile memory; raw identities never touch the external network.
- Agent 2 (Sovereign Dispatcher): Negotiates deployment models across Azure AI Foundry nodes, routing exclusively to compliant sovereign regions (e.g., UK South).
- Agent 3 (Reconstructor & Audit): Ingests the model inference output, seamlessly restores real values, and writes a tamper-evident audit log.

Streamlit Prototype Workflow

Side-by-Side Zero-Friction Governance

- Raw Input: Master Services Agreement with £250,000 fee and London legal jurisdiction.
- Sentinel Redaction: Real-time substitution into {{ORG_1}}, {{ORG_2}}, and {{MONEY_1}} with UK_ONLY tier policy tagging.
- Foundry Execution: gpt-5-mini in UK South processes reasoning entirely over synthetic tokens with zero access to proprietary identities.
- Final Delivery: Fully restored, professional summary delivered to the user with a real-time compliance audit log.

JUDGING CRITERIA

Why SovereignGate Wins

Evaluation Against Microsoft Agentathon Standards

- Innovation: Shifts DLP from static blocking to an active, autonomous multi-agent proxy with real-time token re-inflation.
- Usability: Fully native to Azure AI Foundry SDK. End users write standard prompts without learning specialized compliance syntax.
- Impact: Immediate elimination of GDPR fines (up to 4% global turnover) and zero intellectual property leakage.
- Vendor Agility: Open architecture built on Python that extends naturally to browser extensions, M365 Copilot, and external CLMs.

LOOKING FORWARD

Roadmap & Scaling

Production Hardening over Post-Agentathon Timeline

- Vectorized Policy Knowledge Base: Azure AI Search integration for live clause checking against evolving EU/UK statutory guidelines.
- Browser Extension & Proxy Mode: Zero-click enterprise gateway automatically proxying any corporate AI request.
- Cosmos DB Audit Ledger: Long-term encrypted audit persistence for ISO 27001 and SOC 2 type II compliance verification.
- Automated Red-Teaming: Continuous evaluation using Azure AI Foundry safety evaluators to guarantee zero token leakage.